High-Medium-Low Energy pumps categorization

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1. Purpose

The document describes the criteria used to categorize the pumps according to the energy level of the handled fluid when operating at rated point.

The spreadsheet on Mod.469 calculates the level of energy achieved by a centrifugal pump at specified operative conditions.

Pump with high energy level requires an advanced design to minimize any source of vibration or structural stress on the pump, i.e. the reduction of the pressure pulsations.

2. Scope

This procedure is applicable to all the centrifugal pumps.

3. Abbreviations

Eta	*[%]*	*pump efficiency*
g	*[m/s^2 or ft/s^2]*	*acceleration of gravity*
Hbep	*[m or ft]*	*pump head at best efficiency point and maximum diameter*
N	*[rpm]*	*pump rotor speed*
Nq or Ns	*[/]*	*specific speed of the pump*
Nst	*[/]*	*number of stages*
Nsuct	*[/]*	*number of impeller suction sides*
POWER	*[kW or HP]*	*pump shaft power*
Qbep	*[m3/h or GPM]*	*pump flow rate at best efficiency point and maximum diameter*
SG	*[/]*	*specific gravity*
U2	*[m/s or ft/s]*	*peripheral velocity of the impeller at maximum diameter*

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4. General procedure: description

The procedure classifies the energy level of a selected pump running on specified operating conditions.

The procedure is based on the evaluation of energy level, using different approaches:

- Gulich "Centrifugal Pumps" [1]
- Energy level classification for API610 12 ed preview [2]
- API610 Std 11 ed [3]
- ARAMCO Std 2020 ed [4]
- Internal procedure of TPI [5]

The procedure calculates the values of Head per stage, peripheral velocity and power for low-energy, mid-energy and high-energy pump with the same specific speed of the input pump. Successively the energy level of the pumps is defined comparing the performances with the values previously calculated.

For sake of simplicity all the described methods are expressed in metric unit. The Mod.469 tool provide the correct unit conversion in case of need.

A description of each approach for the evaluation of the level of energy in described below.

4.1. Gulich approach [1]

The Gulich approach defines three energy classes (G1 - G2 - G3) based on the energy level reached by the pump for different values of specific speed. Gulich [1] defines several charts where the energy classes are correlated to:

- the Head per stage at bep multiplied for the Specific gravity of the pumped fluid.
- the peripheral velocity of the pump at maximum diameter and nominal speed.
- the power absorbed by the pump.

The curve in Figure 1 shows the energy classes as function of specific speed of the pump and Hst*SG. In the graph, G3 corresponds to the low energy curve, G2 to the High energy curve, G1 curve is correlated to the ultra-energy curve and is out of scope of work of this procedure.

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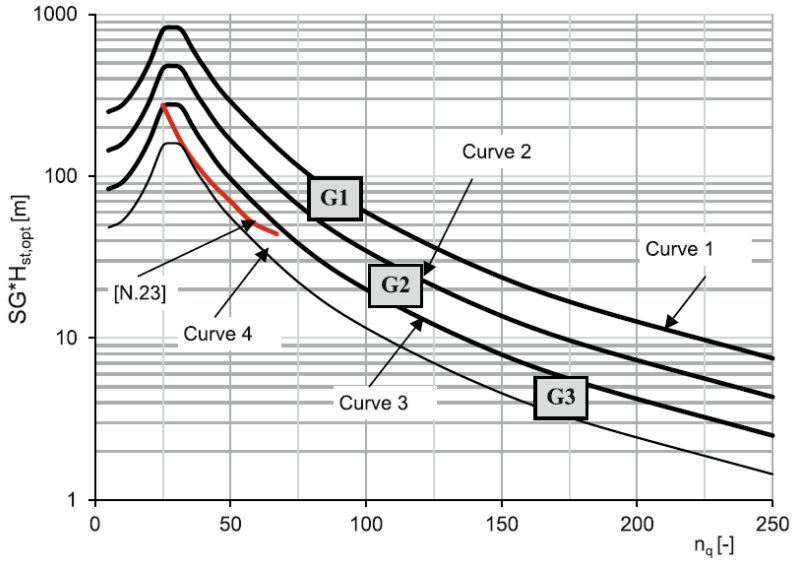

*Figure 1. energy classes as function of specific speed and SG*Hstage.[1]*

On Figure 2 the energy classes are correlated to the specific speed and the peripheral velocity of the impeller at maximum diameter.

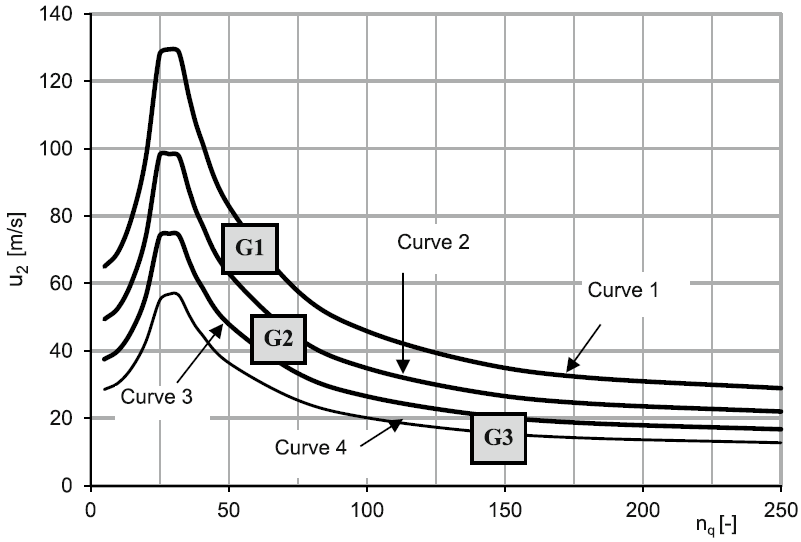

Figure 2. energy classes as function of specific speed and U2.[1]

Figure 3.shows a correlation between the specific speed of the pump and the useful power (Q*H*g*density).

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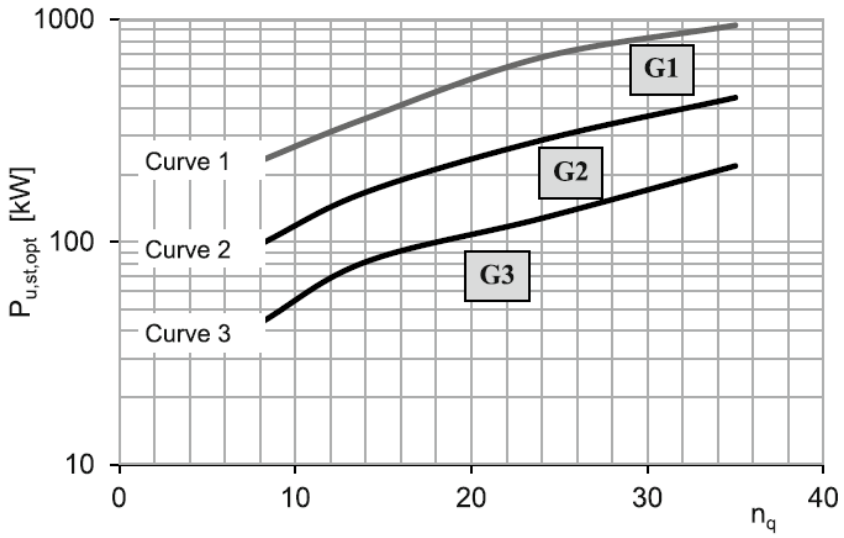

Figure 3. Energy classes definition as function of specific speed and useful power.[1]

The curves in the graph have been refitted with splines. The coefficients of the splines are listed in Table 1 , Table 2 and Table 3.

*Table 1. Coefficients for Hst*SG energy classes splines*

Nq range	coefficient	G1	G2	G3	Formula
Nq<26	C0	196	94.4	55.6	Hst.=C0+C1*Nq+C2*Nq^2+C3*Nq^3
	C1	18.19048	15.43333	9.342857	
	C2	-1.92857	-1.5	-0.88143	
	C3	0.086667	0.058667	0.034	
26<Nq<32	C0	800	460	270	Hst.=C0
Nq>32	C0	2111222	1256659	713901.9	Hst.=C0*Nq^EXP
	EXP	-2.27154	-2.27889	-2.27322	

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Table 2. Coefficients for U2 energy classes splines

Nq range	coefficient	G1	G2	G3	formula
Nq<26	C0	60.4	43.6	34.6	U2.=C0+C1*Nq+C2*Nq^2+C3*Nq^3
	C1	1.2	1.861904762	1.019047919	
	C2	-0.09	-0.154285714	-0.092857143	
	C3	0.006	0.006666667	0.004666667	
26<Nq<32	C0	128	98	75	U2=C0
Nq>32	C0	327.6915363	249.7826123	192.9646977	U2=C0+C1*Nq+C2*Nq^2+C3*Nq^3+C4 *Nq^4+C5*Nq^5+C6*Nq^6
	C1	-10.28604931	-7.803731017	-6.115721035	
	C2	0.165446675	0.125432107	0.099579512	
	C3	-0.001458831	-0.001110805	-0.000889394	
	C4	0.000007161	0.000005496	0.000004416	
	C5	-1.83E-08	-1.42E-08	-1.14E-08	
	C6	1.90E-11	1.49E-11	1.19E-11	

Table 3. Coefficients for Useful Power energy classes splines

Nq range	coefficient	G1	G2	G3	formula
all	C0	-373.92	-29.056	73.245	Pu=C0+C1*Nq+C2*Nq^2+C3*Nq^3
	C1	61.216	17.134	-0.7391	
	C2	-0.8756	-0.3191	0.0905	
	C3	0.0056	0.0063	0.0014	

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4.2. Paper TPS 2014: API 610 12th ed. Preview [2]

This approach establishes the limit between high and low energy pumps, based on the diagram presented on Turbine and Pump Symposium 2014, Houston for the preview of API 610, 12th edition [2].

The chart, shown in Figure 4, correlates the total pressure rise per stage with the specific speed. The semiempirical method is based on several pumps designed for rocket, boiler feed and others industrial applications.

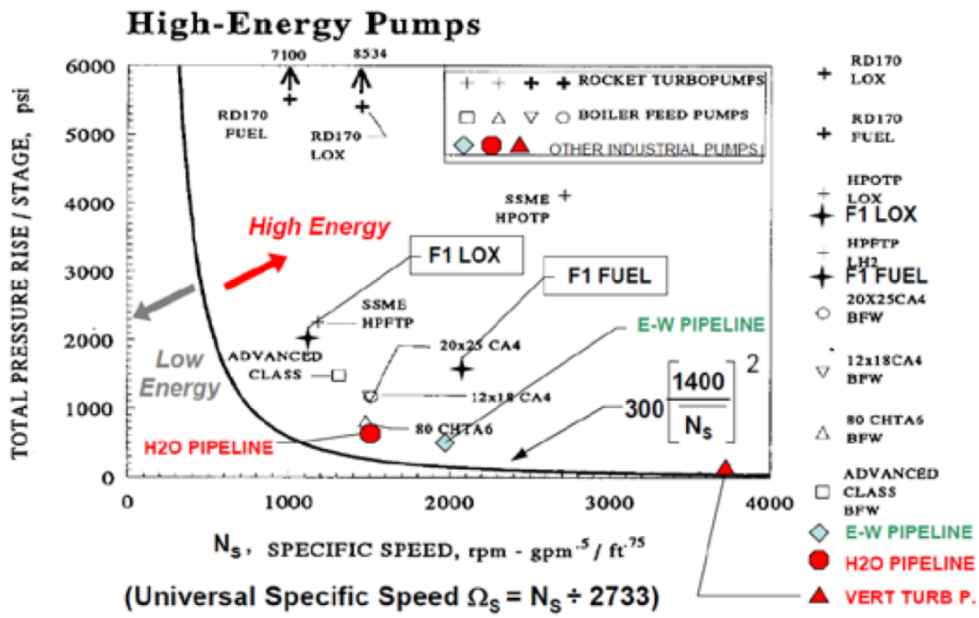

Figure 4 Total Pressure definition as function of specific speed (Us units) [2]

The coefficient used on refitting the curve in the chart, using head per stage and specific speed in metric units, are shown in Table 4. This method is limited to Ns =4000

Table 4. Coefficients for Head per stage API12th ed splines

Nq range	coefficient	G1	G2	G3	formula
<77 [NS=4000]	C0		210.97488		Hst.=C0*(C1/Nq)^2
	C1		27.110767		

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4.3. API 610 11th edition [3]

The approach described in this session is based on the definition of High energy pumps proposed by API 610 11th edition par. 6.1.15 [3] and shown in Figure 5.

6.1.15 Pumps with heads greater than 200 m (650 ft) per stage and with more than 225 kW (300 hp) per stage shall be deemed high-energy pumps and can require special provisions to reduce vane passing-frequency vibration and low-frequency vibration at reduced flowrates. For these pumps, the radial clearance between the diffuser vane or volute tongue (cutwater) and the periphery of the impeller blade shall be at least 3 % of the maximum impeller blade-tip radius for diffuser designs and at least 6 % of the maximum blade-tip radius for volute designs. The maximum impeller blade-tip radius is the radius of the largest impeller that can be used within the pump casing (see 6.1.4). The clearance, P, expressed as a percentage, is calculated as given in Equation (1):

Figure 5. definition of high energy pump – API 610 11th ed. [3]

The limit for High Energy pump definition is Head per stage greater than 200m [650 ft] and power per stage higher than 225 kW [300HP]. Be aware that both the expressions must be verified, otherwise the pump can be considered as mid or low energy.

4.4. ARAMCO Std SAMSS 2023 [4]

The high energy classification based on Material System Specification by ARAMCO [4] is summarized as shown in Figure 6. As per API 610 11th ed pumps with energy higher than 224 kW [300HP] must be considered as High Energy. No limitation has been defined for the head, so it defines a higher restriction on the evaluation of high energy level

This formulation is applicable only if the customer specifically expresses the pump design to be in accordance with the ARAMCO std [4]

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6.1.20 (Exception) For all pumps excluding Helico-Axial Multiphase pumps, the radial clearance between the diffuser vane or volute tongue (cut water) and the periphery of the maximum impeller blade shall be per the below table:

Rated Power per Stage	Volute Design Casing (% of max impeller radius)	Diffuser Design Casing (% of max impeller radius)
≤ 224 KW (300 HP)	6	3
>224 KW (300 HP)	8	4

Note: The vendor can limit the value for the maximum impeller diameter specified in the datasheet to meet this requirement as long as the vendor still meet the mandatory 5% additional head mandated in 6.1.5.

Figure 6. high energy definition – Aramco std [4]

4.5. Internal procedure for Impeller to casing gap example- HighEnergy

The approach defines the zones of high and low energy based on the chart shown in Figure 7. The diagram correlates the power per stage with the head per stages of centrifugal pumps.

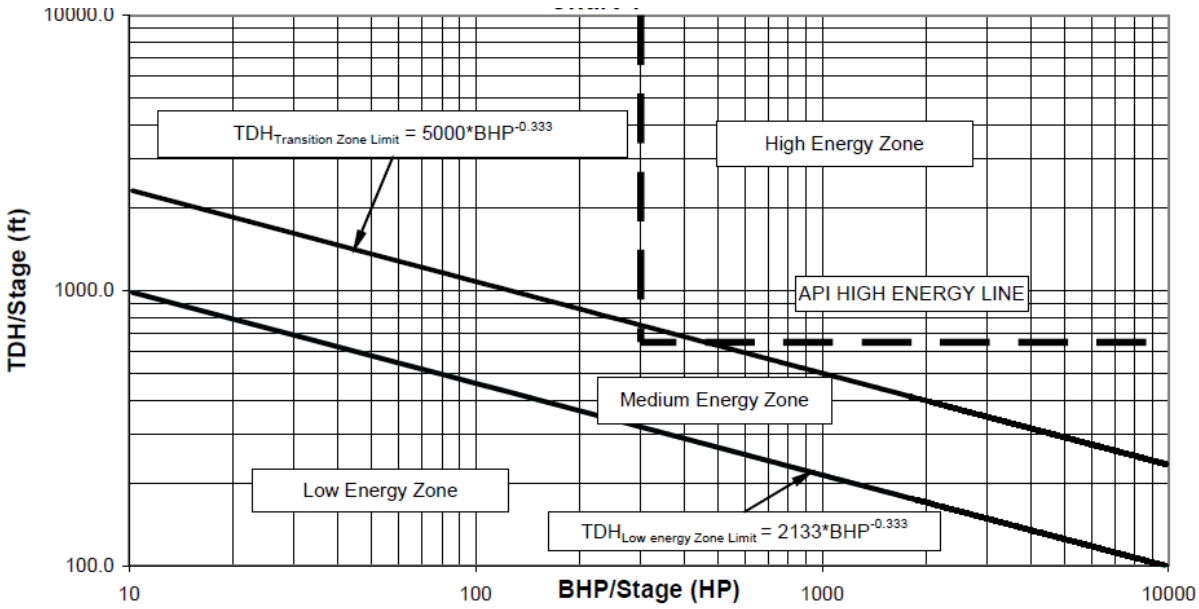

Figure 7. Energy level definition by internal procedure

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Two transition zones are defined to separate low to mid and mid to high energy operation. The dashed lines give the zones of high energy level defined by API 610.

The coefficient used to define the transition zones are shown in Table 5

Table 5. Coefficients for head per stage splines from internal procedure

Nq range	coefficient	G1	G2	G3	formula
all	C0		1400	595	Hst.=C0*Power.st^EXP
	EXP		-0.333	-0.333	

4.6. Energy categorization

All the above-mentioned approaches will be used to identify the energy level of each stage within the pump. The energy classification is based on the rules shown in Table 6

Table 6. Energy classification criteria

	Gulich (chapt.4.1)	Paper TPS 2014 (chapt 4.2)	API 610 11th ed. (chapt.4.3)	ARAMCO SAMSS std. (chapt.4.4)	Internal ST.WK. (chapt.4.5)
Categorization criteria	U2 OR H.stage OR P.stage OR	H.stage	H.stage P.stage	Power.stage	H.stage
LEVEL for High Energy classification (fulfilled if at least one correlation in this row is satisfied)	>G2	>G2	Hst>200m (656ft) AND Pst>225kW (300HP)	>225 kW (300HP)	>G2
LEVEL for Mid Energy classification (fulfilled if at least one correlation in this row is satisfied)	G2 to G3	G2 to G3	Hst>200m (656ft) OR Pst>225kW (300HP)	<225 kW (300HP)	G2 to G3
LEVEL for Low Energy	Whenever is lower than above				

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5. Description of the calculation tool

The tool can be used on both SI or US units.

The list of input and output data of the Mod.469 tool are described below.

5.1. Input data

The input data of the tool are

- the performance curve of the pump with rated fluid (n, Q, H, Power, Efficiency)
- the number of stages Nst and the number of suction sides Nsuct
- External maximum diameter of the pump
- ARAMCO std application [4]

The mask of the input data is shown in Figure 8

INPUT DATA

	unit system			SI	
	Pump speed			2950	RPM
	Q	Hpump	Power	ETA	D2
	m3h	m	kW	%	mm
BEP	10000	380	5000	88	650
	customer request of ARAMCO std. ?			n	

pump name	MOL10000.380	
number of stages	1	
number of suction	2	
Q/side	5000	m3h
H/stage	380	m

Figure 8. Input data.

5.2. Output data

The output data of the tool are

- Specific speed of the pump Nq and Ns
- Peripheral speed of the pump at maximum diameter and rated flow
- Energy level rank based on several methods
- Definition of the energy level of the pump

The rank values are defined as explained in the legend: low energy pump have 0.1 rank, mid energy pump have 0.5 rank and high energy pump have 1 rank.

The definition of the energy level is based on the query of the ranking matrix, using the following rules:

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- If Gulich or API 610 12th ed. Preview OR internal rules OR Aramco (if applied) give rank = 1 for one of the criteria used for the evaluation (Hstage, U2, Power) OR if API 610.11th ed give rank =1 on both Hstage and Power then the pump is classified as HIGH ENERGY
- If Gulich or API 610 12th ed. Preview OR internal rules OR Aramco (if applied) give rank = 0.5 for one of the criteria used for the evaluation (Hstage, U2, Power) and API 610.11th ed give rank =0.1 on both Hstage and Power or Aramco (if applied) gives rank = 0.1 then the pump is classified as MID ENERGY
- The remaining cases are classified as LOW ENERGY

The output data are shown in Figure 9:

OUTPUT DATA

LEGEND	En.lev.	Low	Mid	High
	Rank	0.1	0.5	1
	Range	<G3	G3-G2	>G2

Nq	40.4
Ns	2086.0
U2	100.4

RANK	Gulich	API.610-12th ed	API.610 11th ed	internal	ARAMCO
Hstage	0.5	1	1	1	
U2	1				
Power	1		1		
tot	1	1	1	1	

CALCULATED ENERGY LEVEL	HIGH

Figure 9. output data table.

In addition, the tool shows different charts showing the pump energy level definition with different approaches as shown in Figure 10

G1 line refers to the limit between low energy and mid energy pump, G2 line is the limit between mid and high energy pump.

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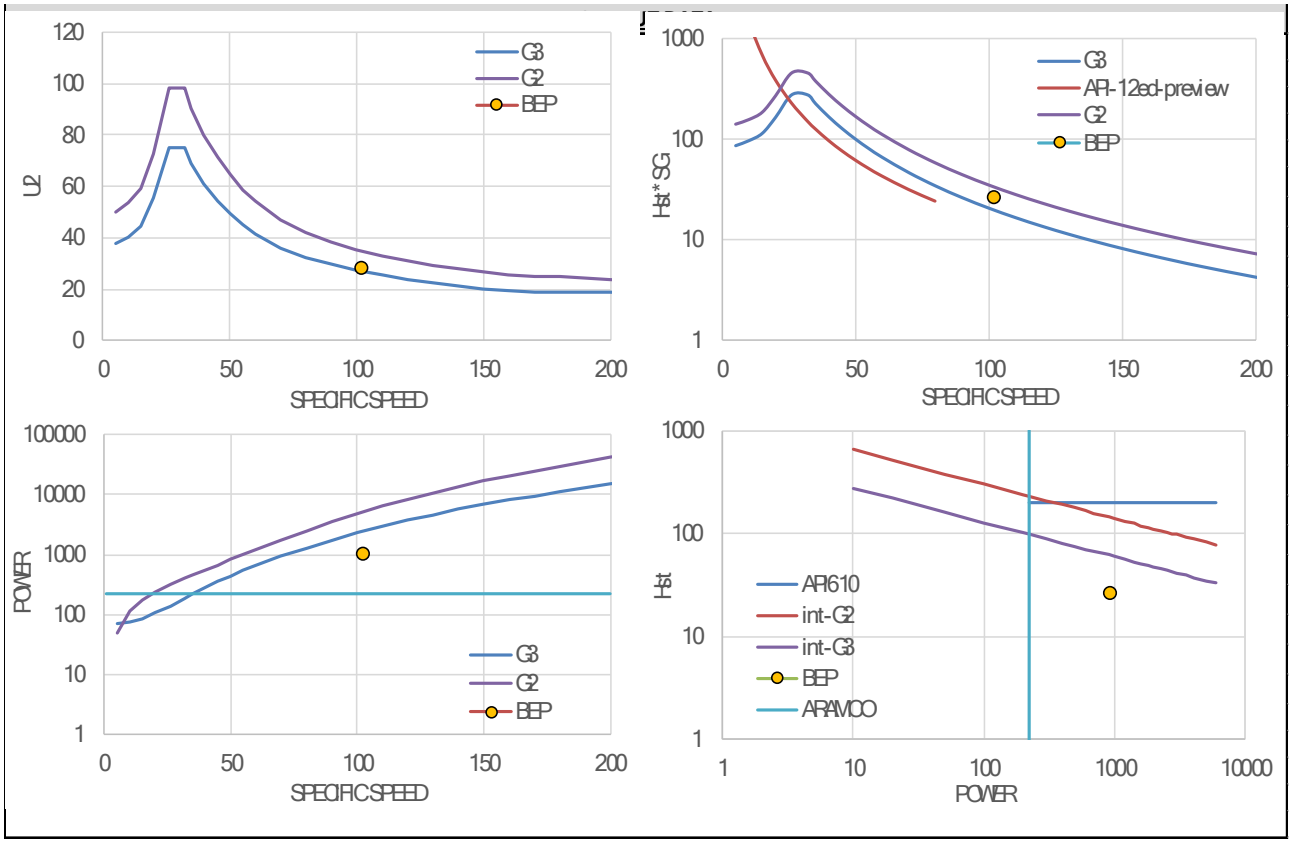

Figure 10. output data - charts

6. Design recommendations

The method described in the document is used to categorize the energy level of the pump.

Whenever a pump is classified as high energy by this tool, the design of the pump must be considered as special and cannot be treated as standard product. For this reason, it's necessary to contact engineering or R&D departments to adjust the pump design in order to avoid critical failure during operation.

For high specific speed, even if the head is low, the power of the pump tend to be high due to the high flowrate, so the pump should be considered as High Energy. Because of this the pump fall in the customer product, so further investigation by Engineering or R&D department must be conducted during the design phase.

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7. References

[1]- Gulich "Centrifugal Pumps" ,4th.ed.

[2]- API 610 12th Edition addresses pump reliability, maintainability and more – TPS 2014 Houston

[3]- API 610 11th ed.

[4]- Material System Specification, 31-SAMSS-004, ARAMCO std -2023

[5]- Internal Procedure

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8. Annex

8.1. Case study: BB1 pump type MOL10000.380 (PROJECT 30598.01)

The pump was designed for main oil line applications. 2950rpm, q=10000m3/h, head=380m, Power=12MW, single stage, double suction BB1 type pump. The tool identifies the pump as HIGH ENERGY (Figure 11)

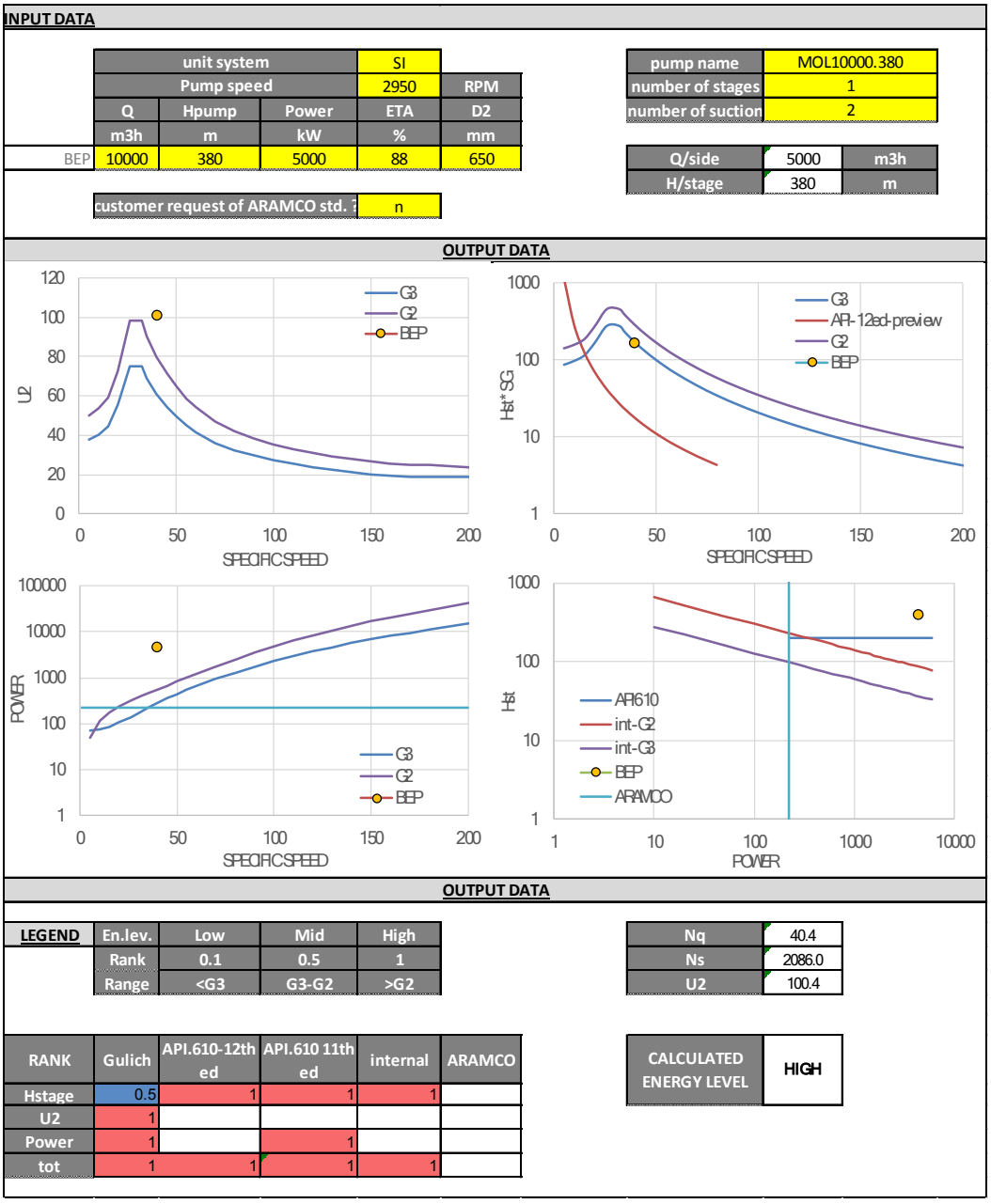

Figure 11. High Energy pump – MOL10000.380.

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8.2. Case study :VS1 pump type 85C1PPA95 (PROJECT P30051.01)

Vertical turbine pump VS1 type 595rpm Q=13756m3/h H=25.4m Nq103. The tool identifies the pump as MID ENERGY (Figure 12)

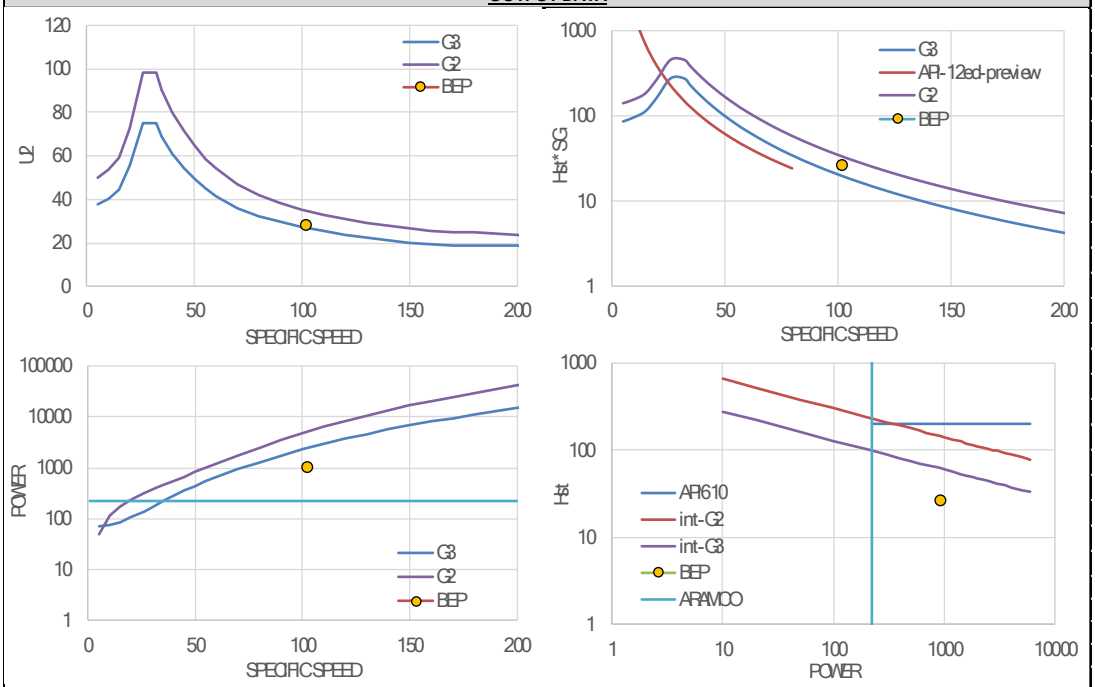
INPUT DATA

	unit system			SI	
	Pump speed			595	RPM
	Q	Hpump	Power	ETA	D2
	m3h	m	kW	%	mm
BEP	13756	25.4	1225.2	77.6	892

customer request of ARAMCO std. 7	n

pump name	85C1PP95	
number of stages	1	
number of suction	1	
Q/side	13756	m3h
H/stage	25.4	m

OUTPUT DATA



OUTPUT DATA

LEGEND	En.lev.	Low	Mid	High
	Rank	0.1	0.5	1
	Range	<G3	G3-G2	>G2

Nq	102.8
Ns	5308.5
U2	27.8

RANK	Gulich	API.610-12th ed	API.610 11th ed	internal	ARAMCO
Hstage	0.5	0.1	0.1	0.1	
U2	0.5				
Power	0.1		1		
tot	0.5	0.1	0.1	0.1	

CALCULATED ENERGY LEVEL	MID

Figure 12.Mid Energy pump -85C1PPA95

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